

12.1 Presentations from R Markdown

Knit slides from code. ioslides, slidy, beamer, or Quarto reveal.

1. Why knit slides from code

A slide deck that is also a source file has one job: the numbers and plots on the slide come from the same code you would put in a homework file. Knit (or Render) again after a data fix and the deck updates. You are not pasting a screenshot of a plot you can no longer reproduce.

The file is still Markdown plus chunks. The output is slides instead of a scrolling report. Note 1.5 is the document system; this note is the slide formats.

2. Formats

R Markdown presentation outputs you will see:

Output	What you get
<code>ioslides_presentation</code>	HTML slides (the usual starting point)
<code>slidy_presentation</code>	HTML slides, W3C Slidy
<code>beamer_presentation</code>	PDF slides. Needs LaTeX
<code>powerpoint_presentation</code>	A <code>.pptx</code> file
<code>xaringan::moon_reader</code>	HTML slides (remark.js). Needs the <code>xaringan</code> package

Quarto's HTML slide format is `format: revealjs`. Lab 12 accepts that as well.

Pick **one** HTML format and stay with it for the lab. Beamer is for when you already have LaTeX and want a PDF. PowerPoint is for a file someone else will click through in PowerPoint.

3. YAML and slide breaks

R Markdown uses `output:`. A minimal ioslides header:

```
---
title: "Arrival delay"
author: "Your Name"
output: ioslides_presentation
---
```

The title slide is built from `title` and `author`. Every later slide starts with `##`.

```
## A slide title

A sentence.

```{r}
mean(mtcars$mpg)
```
```

is a section (ioslides can turn it into a divider). The slides you write are ##.

xaringan is the exception: slides are separated by a line with only ---, not by ##. If you pick xaringan, follow a xaringan template. If you pick ioslides or slidy, use ##.

Quarto reveal uses format: revealjs and the same ## slide titles.

4. What to show

Do **not** show package-load code. Put library() in a setup chunk with echo = FALSE (R Markdown) or #| echo: false / #| include: false (Quarto). The reader came for the table and the plot.

```
```{r setup, include=FALSE}
library(tidyverse)
```
```

A chunk that **computes** a table can show the code (echo = TRUE). A chunk that **draws** the plot the audience should look at often hides the code (echo = FALSE). Both chunks still run.

HTML widgets (an interactive plotly chart, a DT table) need an **HTML** format: ioslides, slidy, xaringan, or Quarto reveal. They will not appear in Beamer or PowerPoint.

PowerPoint is one-way. Knit writes a .pptx. Edits you then make in PowerPoint do not come back into the .Rmd. If you need to change a number, change the source and knit again.

5. One idea per slide

A slide is not a report page. One claim, one table, or one plot. If you need a second plot, it is a second slide.

A few habits that keep a short deck readable:

- title the slide with the claim, not the function name
 - show the code only when the code is the point
 - give the plot room; drop theme_bw() legends you do not need
 - end with a sentence the audience can repeat
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Starter `.Rmd` files are in the Data / templates list on the course page: ioslides, slidy, Beamer, PowerPoint, xaringan, and Shiny. Download one and knit it. Do not vendor a 10 MB knitted deck.

xaringan uses `---` between slides, not `##`. RStudio: **File** → **New File** → **R Markdown** → **From Template** → **Ninja Presentation**. `{xaringanExtra}` adds tile view (press `o`). `{xaringanthemer}` sets colors. `{plotly}` and `{Diagrammer}` are HTML widgets: they need an HTML format.

A Shiny presentation is not a static file. The viewer needs R running. Use it for a live demo, not for a PDF you email.

`{revealjs}` and `{rmdshower}` are other HTML decks. Quarto `revealjs` is the current default if you are starting fresh.

6. Practice

Outline a 3-slide deck (do not spend the hour polishing fonts):

1. Title slide (YAML `title / author`).
2. A slide with a small table (a `summarize()` of a built-in table, or `mtcars / penguins`).
3. A slide with a plot and **no** code (`echo = FALSE`).